COMSATS University Islamabad
Department of Computer Science
Course Description Form (CDF)

Course Information

Course Code: **CSC262** Course Title: **Artificial Intelligence**

Credit Hours: **3(2,1)** Lecture Hours/Week: **2**

Lab Hours/Week: **3** Pre-Requisites: **None**

Course Objective

- To introduce students to the fundamental concepts, history, components, and architectures of Artificial Intelligence, including intelligent agents and their environments.
- To equip students with problem-solving and search techniques, including uninformed, informed, and adversarial search strategies for intelligent decision-making.
- To develop an understanding of knowledge representation, expert systems, constraint satisfaction, and reasoning under uncertainty in AI applications.
- To encourage ethical awareness and critical thinking regarding the societal and philosophical implications of AI, and to explore future directions in the field.

Course Content

This course gives a broad overview of the fundamental theories and techniques of Artificial Intelligence. Topics include: Artificial Intelligence Foundation; Problem-Solving; Adversarial Search; Constraint Satisfaction Problems; Knowledge Representation & Reasoning; Uncertainty and Probabilistic Reasoning; AI Domains Overview; AI Ethics and Future Trends.

Unit wise Major Topics

Unit	Topic	No. of Teaching Hours
1	Artificial Intelligence Foundation: Concepts, History, AI applications; and Intelligent agents & Environments.	4
2	Problem-Solving: Problem-Solving Agents; Searching: Search Algorithms, Uninformed & Informed Search Strategies, Local Search & Optimization Problems, and Heuristic Functions.	4
3	Adversarial Search: Game Theory, Heuristic, Min-Max Procedure & Alpha-Beta Pruning; and Monte Carlo Simulation.	4
4	Constraint Satisfaction Problems (CSPs): Defining, Constraint Propagation, Inference, and Backtracking.	4
5	Knowledge Representation & Reasoning: Knowledge-Based Agents, KR using First-Order Logic(FoL), Inference in FOL; Forward & Backward Chaining; Ontologies & Semantic Networks	4
6	Uncertainty and Probabilistic Reasoning: Bayesian Networks(BNs); Inference in BNs; Dynamic Bayesian Networks; Hidden Markov Models (HMMs).	4
7	AI Domains Overview- Basics of Machine Learning (supervised, unsupervised, reinforcement learning);Basics of Natural Language Processing (Syntax, Semantics, Applications)- Basics of Computer Vision (Image Processing, Object Detection)- Basics of Robotics (Perception, Motion, Planning); AI Ethics and Future Trends	6
Total Contact Hours		**30**

Mapping of CLOs and GAs

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	GA
CLO's for Theory				
CLO-1	1	Describe the foundational concepts of Artificial Intelligence and the models of intelligent agents.	*Understanding*	2
CLO-2	2-4	Apply search strategies to solve problems in Artificial Intelligence.	*Applying*	2-4
CLO-3	5-6	Apply knowledge representation and reasoning under uncertainty to solve problems in Artificial Intelligence.	*Applying*	3
CLO-4	7	Apply foundational techniques from key AI domains to solve real-world problems, considering ethical implications.	*Applying*	3
CLO for Lab				
CLO-5	2-6	Implement various AI techniques and algorithms to solve an AI problem.	*Applying*	2-5
CLO-6	1-6	Develop a medium size project using AI techniques in a team environment.	*Creating*	2-6

CLO Assessment Mechanism

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5	CLO-6
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	-	-
Assignments	-	Assignment 1&2 (CCP[1])	Assignment 3 (CCP)	Assignment 4 (CCP)	-	-
Sessional 1 Exam	Sessional 1 Exam		-	-	Lab Sessional 1	-
Sessional 2 Exam	-	Sessional 2 Exam		-	Lab Sessional 2	-
Final Term Exam	Final Term Exam					Lab Project/ Lab Final Term Exam (CCP+CCA[2])

1. CCP : Complex Computing Problem 2: CCA: Complex Computing Activity

Note:

1. Instructor should design theory assignments labeled as Complex Computing Problems (CCP) in such a way that each problem clearly exhibits 2–3 CCP characteristics.
2. Instructor should design theory assignments labeled as Complex Computing Activity (CCA) in such a way that each problem clearly exhibits 2–3 CCA characteristics.

Text and Reference Books

Textbook:

1. Artificial Intelligence: A Modern Approach, Russell, S., and Norvig, P., Pearson, 2020.

Reference Books:

1. Artificial Intelligence Basics: A Non-Technical Introduction, Taulli, T., Apress, 2019.